REQUIREMENT COLLECTION FORM

Note: Read the Guidelines carefully, before submitting.

Name	
Contact	
College Name	
Project Title	Fitness
Base Paper Link	
Expected Inputs	**1️⃣ AI Workout Recommendation Engine** **Input:** ● User profile (age, gender, weight, height, BMI) ● Fitness goals (strength/cardio/flexibility) ● Activity level ● Available equipment **Output:** ● Personalized workout plan ● Exercise type, sets, reps, duration ● Intensity recommendations **Implementation:** ● **Gemini API** (send user data as prompt, get workout plan) **2️⃣ AI Diet Recommendation System** **Input:** ● User goals (weight loss/muscle gain/maintenance) ● Current weight, target weight ● Food preferences (veg/non-veg/vegan) ● Activity level **Output:** ● Daily meal plan ● Calorie targets ● Macro split recommendations **Implementation:**

	• **Gemini API** (send user nutritional needs, get meal plan) **3️⃣ Computer Vision Workout Monitoring MAIN MODULE** **Input:** • Live webcam feed • Selected exercise (push-up/squat/plank) **Output:** • Real-time rep count • Form correctness alerts • Posture feedback **Implementation:** • **MediaPipe Pose** (body landmark detection) • **Rule-based angle calculations** (using landmark coordinates) **4️⃣ Gamification Module** **Input:** • Workout completion data • Daily login tracking • Streak count **Output:** • XP points • Badges unlocked • Leaderboard ranking **Implementation:** • **Rule-based logic** (Python functions) • **SQLite database** for tracking **5️⃣ Nutrition Image Recognition ML MODULE** **Input:** • User-uploaded food image **Output:** • Detected food item name • Estimated calories (via Gemini after detection) • Macro breakdown **Implementation:** • **MobileNetV2** (pre-trained, fine-tuned for food classification) • **Transfer Learning** approach • **Model can be changed during the development if its not performing well** **Recommended Dataset: 🎯 Food-101 Dataset** - https://www.kaggle.com/datasets/dansbecker/food-101 • 101,000 images • 101 food categories • Clean, high-quality images • **Most reliable for training** **Backup Option: Indian Food Images** - https://www.kaggle.com/datasets/iamsouravbanerjee/indian-food-image

	s-dataset (if Indian cuisine focus needed) 7️⃣ **AI Fitness Chatbot** **Input:** • User text query (e.g., "Best exercises for abs?") **Output:** • Contextual fitness advice • Exercise recommendations • Nutritional tips **Implementation:** • **Gemini API** (conversational fitness assistant)
Expected Outputs	
Dataset (If Available)	
Algorithm (If Known)	
Tech Stack: (In case of Full stack project)	**Frontend + Backend** • **Frontend:** Your existing HTML/CSS/JS (unchanged) • **Backend:** Flask (Python) • **APIs:** Gemini API integration • **Database:** SQLite
Deadline	To be decided
Comments	And the project should have the same frontend as i gave before in the zip file . Because i displayed by frontend in my previous review.
Workflow	

Points to Note:

- The project development strictly adheres to the provided algorithm and workflow
- Changes to the above-mentioned entities results in the **entire reconstruction** of the project
- Leave the cells blank if you do not have the necessary data
- Explain everything in detail, requirements that are not mentioned in the form will **cost extra** as per requirement.
- Dataset and algorithms **will not be changed later** once a project development is started